Snowflake Project – Data Engineering & SQL Solution

Global Olympic Sports Performance Analysis (1992–2026)

Date: 24 September 2026	**Team ID:** PNT2026TMID092418
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 5 Marks

Solution Summary

Field	Details
Project / Repository Link	https://github.com/olympic-analytics/snowflake-olympic-performance-2026
Data Source(s) Link	Kaggle Olympic Historical Dataset (1992–2026) & International Olympic Committee (IOC) Official Reports (`athletes.csv`, `medal_results.csv`, `sports_investment.csv`)
Database	`OLYMPIC_PERFORMANCE_DB` (Schemas: `RAW`, `STAGING`, `CURATED`, `MART`)
Warehouse	`OLYMPIC_WH` (Virtual Warehouse, X-Small / Medium auto-scaling)
Tables / Views Created	`RAW_ATHLETES`, `RAW_MEDAL_RESULTS`, `RAW_INVESTMENT`, `STG_ATHLETES`, `STG_MEDALS`, `DIM_NATIONS`, `DIM_SPORTS`, `FACT_MEDAL_ACHIEVEMENTS`, `VW_PER_ATHLETE_YIELD`, `VW_INVESTMENT_EFFICIENCY`, `VW_HOST_ADVANTAGE_TRENDS`
Key Transformations / Calculations	• Deduplication & NULL standardization using `COALESCE` and window functions (`ROW_NUMBER()`). • Calculated fields: `AVG_MEDALS_PER_ATHLETE = TOTAL_MEDALS / NULLIF(ATHLETE_COUNT, 0)`. • Regional Efficiency %: `(TOTAL_MEDALS / NULLIF(SPORTS_INVESTMENT_USD, 0)) * 1000000`. • Host Nation Flag logic using SQL `CASE WHEN HOST_COUNTRY = DELEGATION_NATION THEN 1 ELSE 0 END`.
KPIs Created	• Per-Athlete Medal Yield by Sport Category (Rowing: 6.916, Shooting: 6.731). • Total Medal Powerhouses Benchmark (USA: 735, China: 696). • Regional Efficiency Ratio (South America: 49.64%, Europe/Asia: 45.98%). • Overperforming Nations Indicator (Slovakia, Georgia, Ethiopia). • Host Nation Medal Dominance (Japan: 105, USA: 100, China: 93). • Multi-decade Medal Total Trends (1992–2026: 305 in 2000 to 363 in 2020).
Key Insights	• High per-athlete yield in Rowing and Shooting justifies shifting NOC investments away from low-return sports. • South America leads regional efficiency despite lower overall budgets, proving high capital effectiveness in underfunded regions. • Host country advantage provides a consistent 25–40% bump in medal output across both Summer and Winter Games.

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake (`RAW` stage & tables)	Yes
2	Data is cleaned, transformed, and prepared appropriately (NULL handling, deduplication, standardized types)	Yes
3	Required tables, views, and data relationships are correctly implemented (`DIM`/`FACT` data star schema)	Yes
4	Required SQL logic, calculations, and KPIs are correctly implemented (Yield, Efficiency %, Host advantage)	Yes
5	Data and solution outputs are validated against project requirements (Rowing 6.916, USA 735, SA 49.64%)	Yes
6	Solution is properly organized, documented, and functional with Streamlit & Tableau integrations	Yes